

Analysis of University Alumni Success Rates

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Summary of Research Questions

1. Does acceptance rate meaningfully predict graduate outcomes, such as salary, employment rate?
 - a. We will investigate and compare the acceptance rates between different universities depending on their ranking. Based on these differences, we will compare employment rates and outcomes to determine whether or not higher/lower acceptance rates correspond with varying levels of post-graduate success.
2. How strongly do university rankings correlate with post-graduation average salaries?
 - a. We will investigate and compare different U.S. university rankings to determine whether or not a higher ranking correlates with a higher post-graduate average salary.
3. What ranking factors lead to the highest median salaries in a University?
 - a. We will investigate and compare different university factors, such as faculty/staff qualifications, to determine if there is a correlation between university factors and the salaries post-graduation. This helps us determine what factors of a university correspond to higher monetary success rates within graduates.

Motivation

As university students ourselves, we thought it would be interesting to see how the university that other students attend determines or is correlated to typical measures of success like salary and job industry. University ranking and admissions has been such a topic of obsession for high school students and their parents in recent years, and as acceptance rates for high ranked institutions plummet, it's interesting to see how all this hype measures actual success post graduation.

Dataset

One dataset we will use is: [College Completion and Efficiency Measures for US \(Institution Details\)](#) from “the devastator” on Kaggle. We will be using the “cc_institution_details.csv” file. Scrolling down, there is a download icon to the upper right on the opposite side of the file name “CollegeAdmissions_Data.csv” to download the dataset (downloading requires having an account). This dataset will provide us with information on institutions’ graduation rates, awards, expenditures, and outcomes. From this data, we can examine how factors, such as expenditures, connect with graduation data (rate, length), awards, and more.

A second dataset we will use is: [US Universities & Colleges Dataset \(2025\)](#) from “brahmendra jayaraju” on Kaggle. Downloading the dataset (“universities_data.csv”) requires the same process as the dataset listed above. This dataset contains information such as graduation rate, location, campus setting, and median annual earning of graduates for universities and colleges in 2025. These statistics will be

particularly helpful for us in analyzing measures of success at universities in a recent year. Both datasets provide some demographic information that we could additionally investigate.

A third dataset we will use is: [U.S. Department of Education Scorecard Most recent institutions \(U.S. DOE\)](#) from the U.S. Department of Education. This dataset is accessed through a different source than the kaggle datasets above, and the dataset must be opened in an excel spreadsheet before being saved to a CSV file to avoid missing data. This dataset consists of thousands of columns of statistics for thousands of universities in recent years, but the data that we will prioritize from this dataset are typical measures of educational “success” such as SAT scores and post graduation median incomes.

Access datasets [here](#)

Chosen Challenge Goals

1. Multiple Datasets (To combine databases of alumni with other statistics of universities)
 - a. To achieve this challenge goal, we joined together all the datasets into two combined datasets. These combinations were the [College Completion and Efficiency Measures for US](#) and [U.S. Department of Education Scorecard Most recent institutions](#) and the [US Universities & Colleges Dataset \(2025\)](#) and [U.S. Department of Education Scorecard Most recent institutions](#). This helped us use data from different datasets and variables that other datasets didn't have to combine them and create a better understanding of the correlation between the variables of interest and markers of student success. We didn't end up scaling back and it ended up being a great addition to our analysis.
2. New Libraries (For advanced/interactive visualizations)
 - a. To achieve this challenge goal, we implemented the bokeh library. This helped us better our visualizations and make them more informative through interactive experiences. Since visualizations were especially important for our project in seeing data correlations, we decided this would be the best fit unknown library to use. It was implemented into our combined dataset visualizations. Since our datasets also had thousands of rows of different universities, the bokeh library helps users to pinpoint exact universities on an interactive graph. It also has user tools like plot zooming and isolating select points.

Methods

1. In researching this question, we will aim to webscrape (connected to challenge goals) to gather admissions information. We will want to find acceptance rates, so we can calculate the number of accepted applicants divided by the number of received applicants from our dataset and obtain those percentages. Having acceptance rate values will be beneficial for our analysis of the connection between it and graduation rate and time it took to graduate.
2. We plan to do web scraping (connected to challenge goals) in order to gather rankings for universities since they are not present in the found datasets. Using the datasets we found, we will look at mean salaries of graduates after completing their degree in connection with rankings to compare and contrast.
3. For our third research question, we will compare annual post-grad earnings values from the datasets in connection with the percentage of faculty and student to faculty ratios as metrics for

analyzing how faculty affect students' success at universities. We will also look into the locations and campus settings to investigate connections between environment and post-grad salaries. It will be helpful to organize the data to make efficient comparisons, so we will create dataframes to advance our examinations.

EDA Results

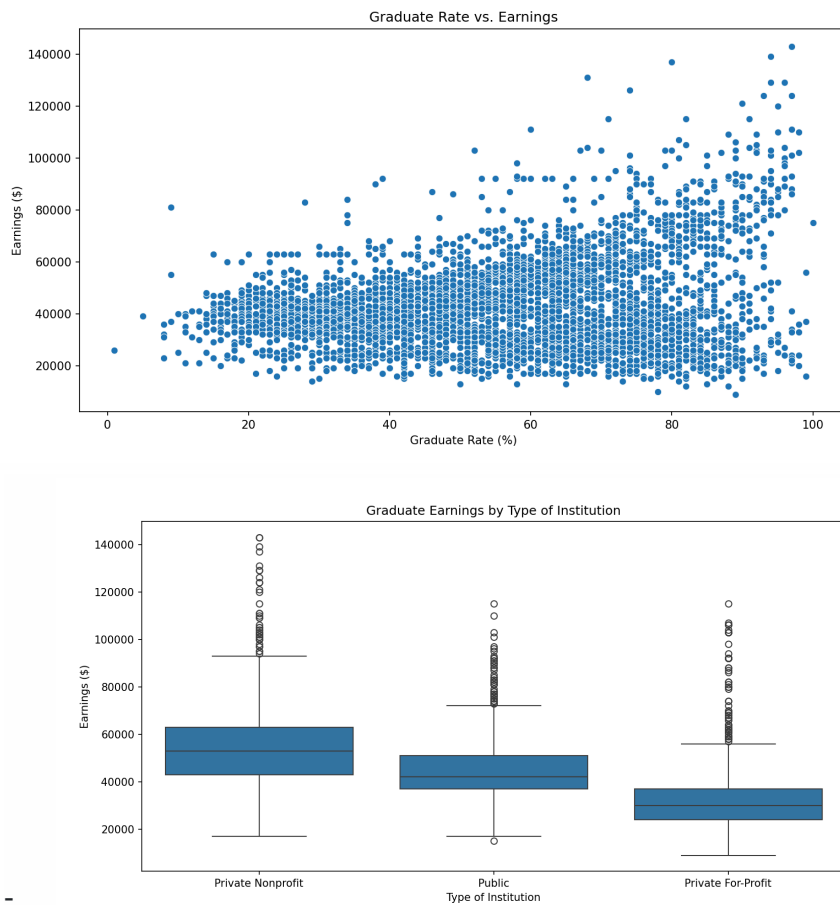
Dataset 1: [US Universities & Colleges Dataset \(2025\)](#)

For this dataset, we examined the size of the data, amount of missing values, and summary statistics for the variables of interest. These were Graduate Rate, Location, Type (private nonprofit, public, or for-profit), Group (large, medium, small school), and the median annual earnings of graduates. These variables were chosen because they were all traits of a school that might affect their National rankings and could be used to determine if there was correlation between school ranking and alumni success.

The analysis of the EDA revealed that there were 5546 different universities in the dataset. Between these schools, graduation rates varied greatly across all the schools with a standard deviation of 21% and a mean graduation rate of 57%. There were also 2563 unique locations of schools, indicating great variability in the spread of schools across the nation. The earnings also showed some variation, ranging from a minimum of \$9,000 and a maximum of \$143,000. The mean of the earnings was \$43,482. One of the visualizations created compared the graduation rate with the earnings of their graduates. The other visualization created was a box plot comparing the earnings of graduates against the type of institution (public, private nonprofit, and private for-profit).

Through this analysis, we discovered that as graduation rate increases toward 100, a student's post graduation median earnings also increases. As well as, private non-profits have a larger distribution and mean but that there isn't a significant difference between the values of each type of institution. However there is private nonprofit has the highest median, public has the in between, and private for-profit has the lowest. This EDA helped us identify that graduation rate has a positive correlation with graduate earnings and that institution type, graduation rates, and earnings are important variables to help us answer whether there is a correlation between university prestige and post-grad student success.

- Graphs:

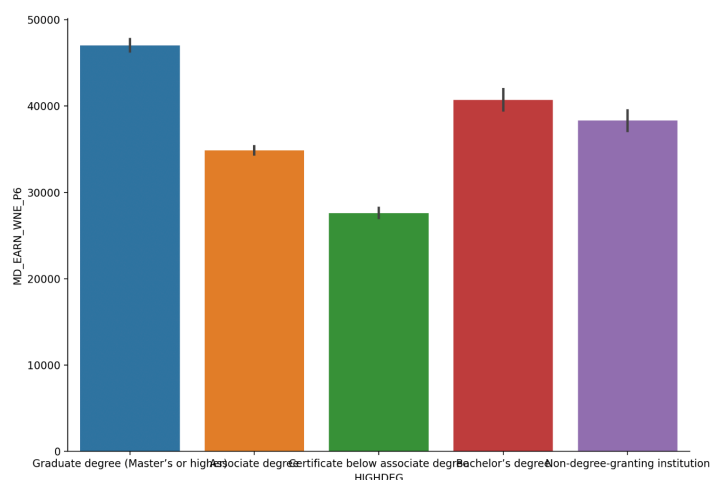
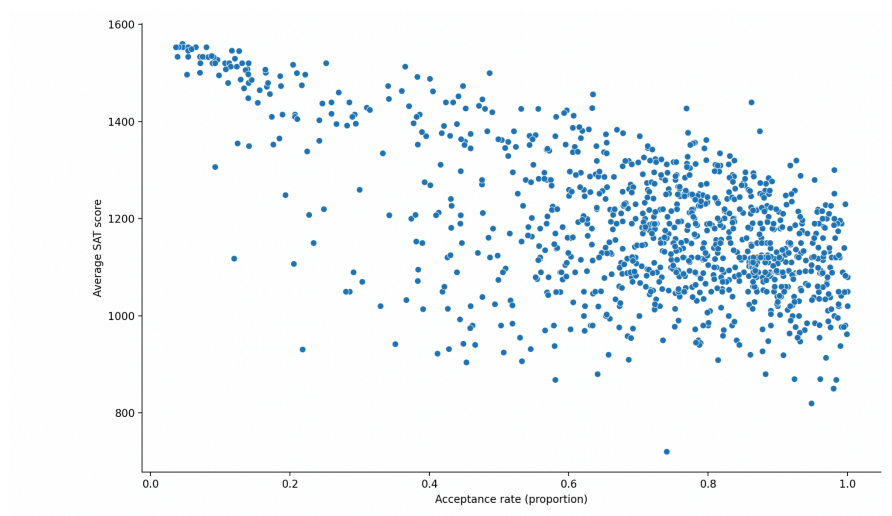


Dataset 2: [U.S. Department of Education Scorecard Most recent institutions](#)

This EDA was evaluated for its variables of interest, which were admission rates, median earnings 6 years post grad, average SAT scores, percentage of students with degrees in computer science, business, and engineering, and the highest degree the school offered. In this dataset, there were 6323 schools being compared. We also evaluated it for missing values and found that there were a significant amount of missing values from the admissions rate and average SAT columns. These missing values were used as a measure of selectiveness, with it helping us understand that many admissions information may be non-selective (community colleges, etc.) or use an alternative admission process.

For the variable summaries, we found that the average sat score was 1180, the earnings had an average of \$37,420. The visualizations created were comparing SAT scores with admissions rates, as well as comparing the highest degree offered with the median earnings of graduates in 6 years. These visualizations revealed a negative relationship between admission rates and average SAT score, suggesting that more selective universities that accept SAT scores have stronger academic backgrounds. The second graph also showed that institutions offering higher degrees tended to have higher median earnings in their graduates. This EDA helped us identify admission rates, SAT scores, highest degrees offered and graduate earnings as variables to support our findings.

Visualizations:



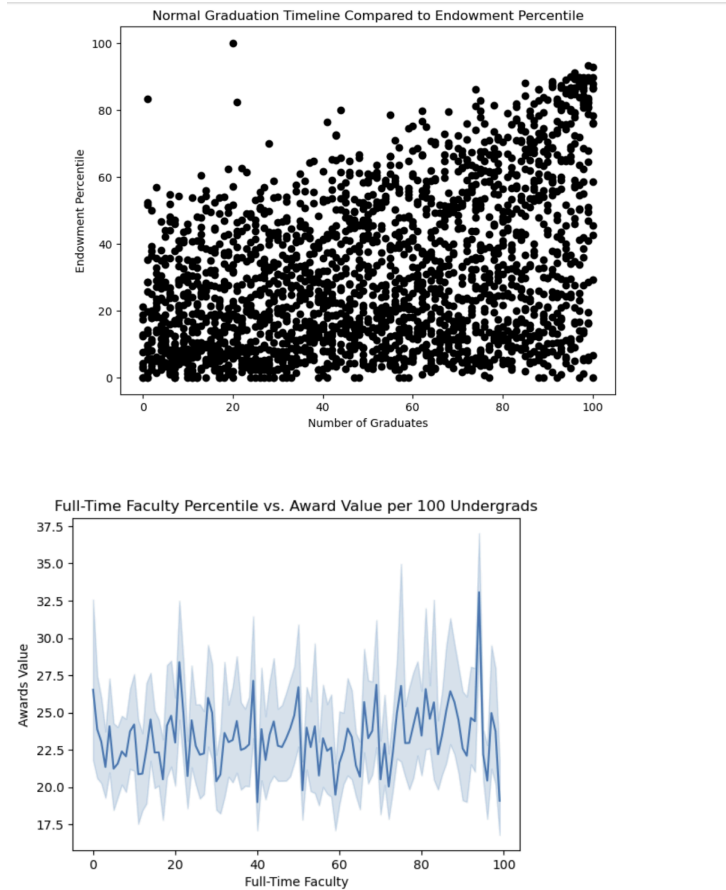
Dataset 3: [College Completion and Efficiency Measures for US](#)

This EDA included 3798 schools and the variables of interest were focused on the amount of awards the school received, the amount of graduates who graduated at the expected time, full time faculty percentile, the percentile of students receiving financial aid, the endowment percentile, percentage of students receiving pell grant, and the median SAT percentile.

The visualizations created compared the number of students who completed their degree in 100% of the normal time frame and the endowment percentile of the school. The other visualization compared the percentage of full-time faculty against the value per 100 undergraduate students of awards given. From these, we can evaluate that there is a slight increase in graduates graduating on time as the endowment percentile increases.

This EDA helped us identify award measures, graduation outcomes, and faculty percentages as factors in determining institution prestige.

Visualizations:



Multiple Dataset EDA

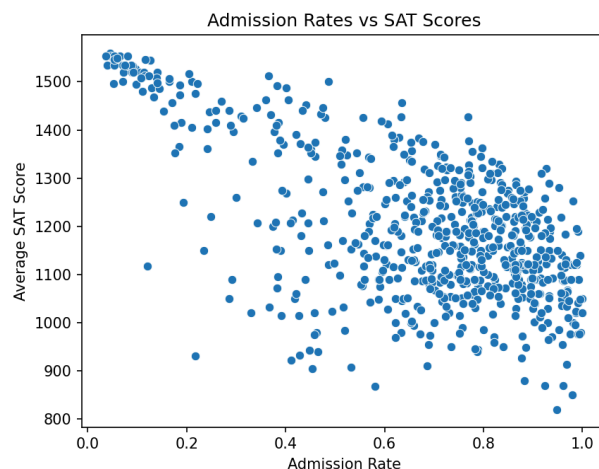
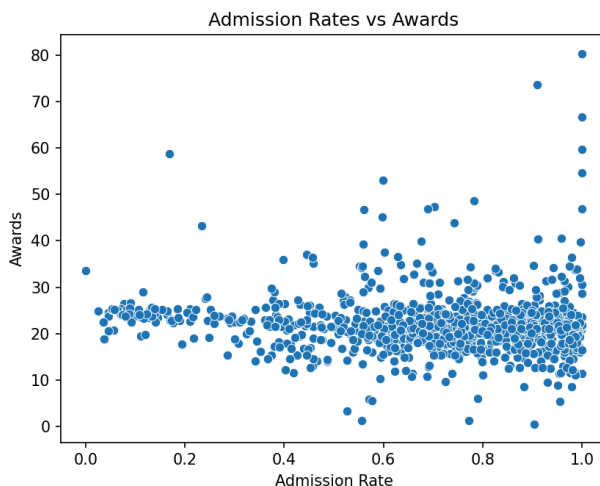
Combination 1: [College Completion and Efficiency Measures for US](#) and [U.S. Department of Education Scorecard Most recent institutions](#)

For this combination, we combined the admission rates from the U.S. Department of Education ScoreCard dataset with the amount of awards of each institution. Through this, we were able to compare the admission rates, awards per value, and SAT average grouped by the same institutions. There were 1904 overlapping schools in these datasets and the merged value had a large amount of missing values in the SAT column, indicating university selectivity.

The admission rates had a mean of 70%, the awards per value had a mean of 21, and the SAT average had a mean of 1195. The visualizations created compared the admission rates to the awards per value, and the admission rates to the SAT scores. The first one revealed that there didn't seem to be a large correlation between admission rates and awards. However, the second graph showed a clear negative relationship between admission rates and average SAT scores, reinforcing the idea that more selective universities require SAT scores and therefore students with stronger academic backgrounds.

This merged dataset helped us evaluate which indicators are more strongly associated with university prestige.

Visualizations:



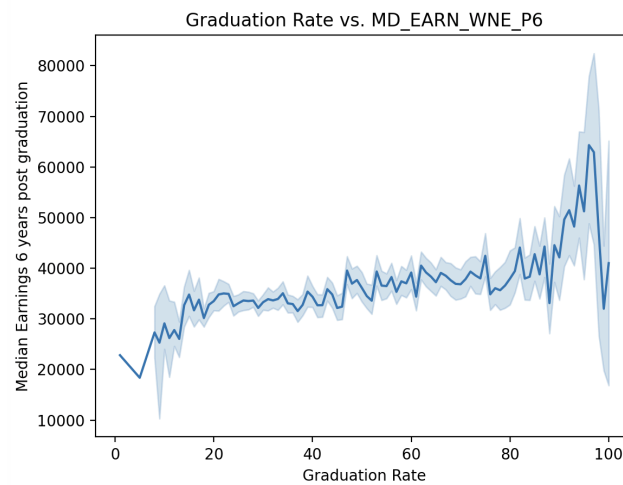
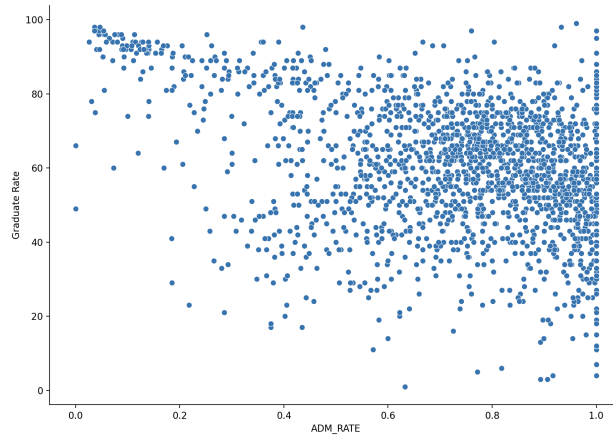
Combination 2: [US Universities & Colleges Dataset \(2025\)](#) and [U.S. Department of Education Scorecard Most recent institutions](#)

This merged dataset combines the acceptance rate from the U.S. Department of Education Scorecard dataset with the graduation rate and earnings from the US Universities and Colleges Dataset. There are 5691 universities in the merged data set. It allows us to directly compare university selectivity with student economic success.

The visualizations created from this dataset compared the admission rates with the graduation rates and the graduate rates with the median earnings 6 years post graduation. From this, it is revealed that as admission rates increase, the graduation rates decrease, but there is a significant amount of more data as admission rates increase as well. It can also be deducted that there is a slight positive correlation between graduation rates and median earnings.

This combined analysis helped connect multiple factors of university success into a single dataset and compare it to the economic success of students from those universities based on the factors given.

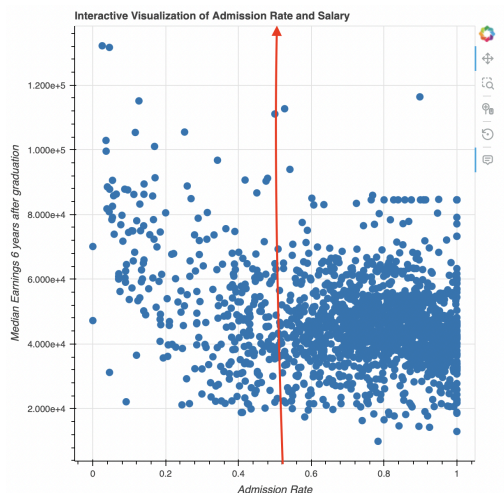
- Visualizations



Results

1. Does acceptance rate meaningfully predict graduate outcomes, such as salary, employment rate?

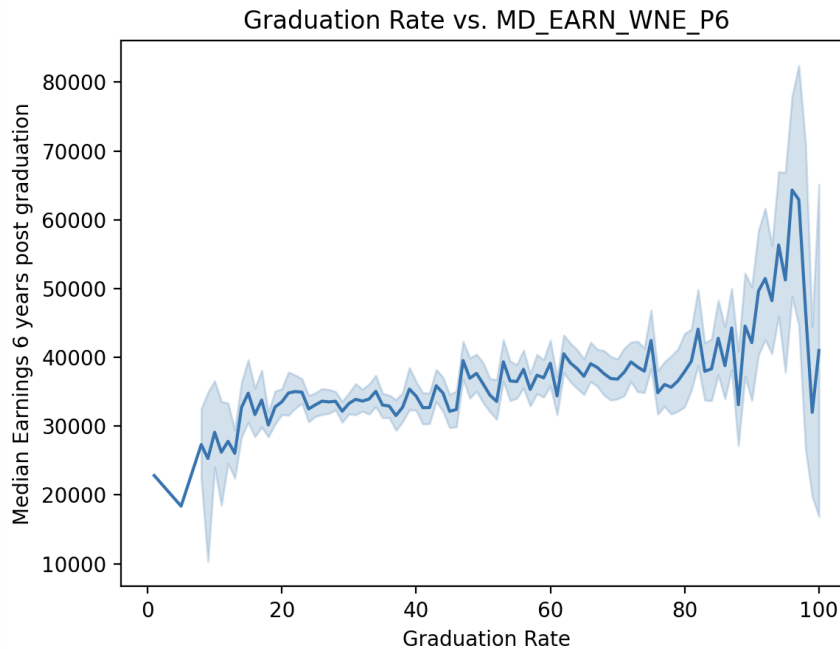
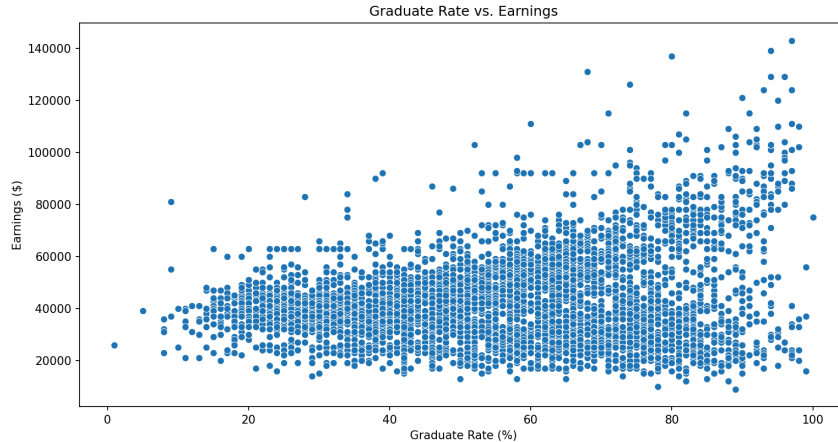
While there's not a strict correlation between acceptance rate and graduate outcomes, there's definitely a slight relationship.



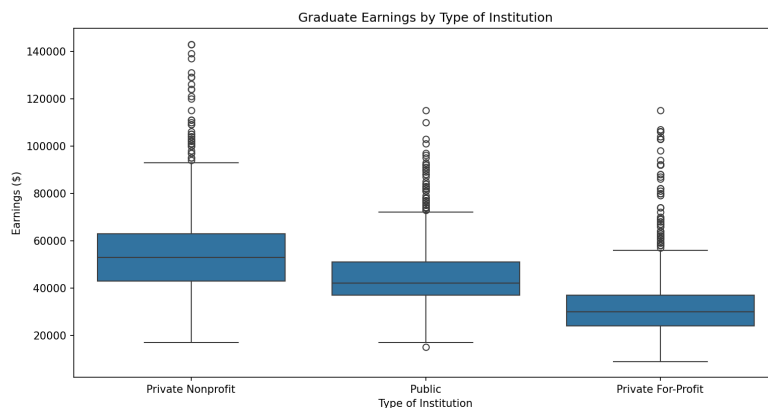
Taking a look at the graph we can see that for around a 50% acceptance rate and above, the salary is pretty similar, apart from some outliers. But after the 50% mark is when there starts to be a real pattern with salaries decreasing as acceptance rate increases. While this finding may not contribute to a definitive prediction that those who go to more prestigious universities determined by a lower acceptance rate have a greater chance of success post graduation, there's definitely a trend upon looking at the evidence.

2. How strongly do university rankings correlate with post-graduation average salaries?

Based on university ranking factors such as graduation rates and the type of institution the university is, we can observe a slight correlation between these factors and the post-graduate earnings.

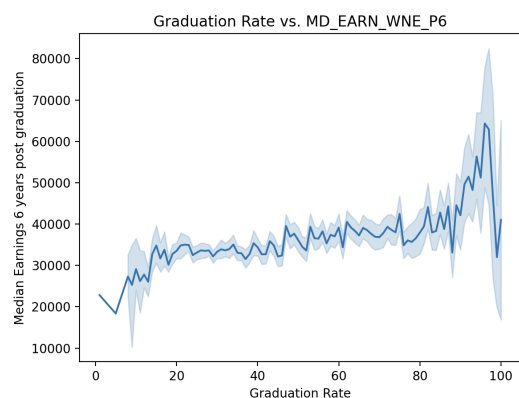


As pictured above, there is a slight increase in the post-graduate earnings as the graduate rate increases. Higher graduation rates indicate a higher university ranking, which indicates that a higher university ranking has a slight positive correlation with higher post-graduate earnings. The first graph comes from the first data set: [US Universities & Colleges Dataset \(2025\)](#) and only contains the data within that dataset. The second graph is from the combined dataset 2. In both these graphs, we can observe an increase in post-graduate salaries as graduation rates increase, which shows that in all different sets of data, it can be concluded that there is a positive correlation between higher university rankings and graduate salaries.



Institution types affect the ranking of universities, as many private universities are usually considered higher ranking. Although this visualization doesn't show a significant amount of correlation between earnings and institution type, we can observe that private universities have a larger spread of graduate earnings, with a higher median and many more outliers with higher earnings compared to public and private for-profit universities. This indicates that there is a slight correlation between the university type and the graduate earnings, with private universities (usually higher ranking) having higher median and outlier graduate earnings.

3. What ranking factors lead to the highest median salaries in a University?



While investigating this question, we had to alter the factors we took into consideration as a result of data and limitations. In our research, we analyzed faculty metrics and graduation data points. With this graph, we can, along with making interpretations related to our second research question, gather that median post-grad salaries increased with the increase in university graduation rate. We took note of the inconsistency at the right-end of the graph and considered an array of explanations for this. There are differences between universities that may affect

graduation rates; graduation rates are not easy to standardize for comparison considering the different goals and levels of institutions. Overall, we see that graduation rates can be a factor in higher median post-grad salaries. Additionally, we concluded that admission rate is a factor that contributes to increases in higher median salaries. One of our visualizations showed us that there was a moderate correlation between admission rate and median earnings six years after graduation. We noted that the amount of data distributed through the graph could contribute to misconceptions, but that there was still a noticeable decrease in median earnings as admission rates increased.

Impact and Limitations

It's extremely important to take all results from this study with a grain of salt especially because university admissions is such a taboo/sensitive topic for millions of students around the world. It's

especially important to highlight that correlation doesn't necessarily deem causation, so any of the results that show that certain measures of university prestige (location, degrees offered, acceptance rate, average SAT etc.) correlate with higher success rate measures (salary) doesn't necessarily mean that success is determined strictly by those factors.

This report could also be potentially harmful to university reputation and biases against particular universities and university students. For example, if a particular student attends a university with a higher acceptance rate than deemed competitive, the associated statistics may lead them to feel bias from outsiders on their student status.

A big limitation is the amount of missing information from some of the datasets since this can lead to biases where we, in this project, are generalizing our findings while there are institutions that are not represented.

On the other hand, some of the results also show many outliers from the typical trends in this study, showing success stories from universities that don't have traditionally competitive statistics.

For students interested in a particular post university stats, this report can be a good guide to help them figure out which universities they want to aim for (especially using the interactive graph with adjustments to the variables that choose).

Work Plan Evaluation

- **Task 1:** Our first task of collecting and cleaning datasets took us longer than predicted. We had trouble finding datasets that had all of the information we were looking for. We constantly had to replace datasets and swap them for others that were more applicable to our project. Once we had sufficient data in .csv format, we were able to combine them and perform computations.
- **Task 2:** Our predicted timeframe for the exploratory data analysis was on-point. Each of us took on one dataset to perform and analyze summary statistics, create visualizations for certain metrics, and document important variables from the datasets and patterns noticed in our analyses.
- **Task 3:** For finalizing research questions, we likely ended up taking the minimum amount of time we predicted. This task largely involved creating graphs to compare variables of the combined datasets and analyzing statistics found from the combined datasets about certain variables like average earnings for post-graduates, graduation rates, acceptance rates, and more.
- **Task 4:** The final task of finalizing the report and reviewing the code required the amount of time we predicted, if not more. Working through code quality improvements to our analysis files, finalizing test files, and organizing all code files was a large objective of this task. We also ensured that our written report was updated with all necessary information without being oversized.
- Our **collaboration plan** was successful. We were able to develop our code individually, then organize it effectively. Each member contributed quality work in a timely manner and clearly communicated any issues that arose with the rest of the team. Before submissions, we ensured that we were all OK on all aspects of the work of the deliverables.

Testing

We created a small test file for each dataset to test the methods we each used to clean and analyze our datasets. In each of these, we created a small mock dataframe that included our variables of interest. From there, we tested each method included in each dataset file with assert statements, making sure that they outputted the right information, such as five number summaries or missing values. Through this, using a smaller dataframe allowed for us to manually input the correct outputs that we would expect in our larger dataframes. These test files tested the datasets because we knew we would end up combining datasets, so we made sure to test our original dataset files to make sure our code was working correctly before implementing the datasets into any new goals, like combining the datasets. After the test files were made, we called them with the original dataset files to ensure that our code was being processed correctly.

Citations and Collaboration

We did not have any collaborations with this project.

Dataset sources:

- [College Completion and Efficiency Measures for US](#)
- [U.S. Department of Education Scorecard Most recent institutions](#)
- [College Completion and Efficiency Measures for US](#)

Research sources:

- <https://robertkelchen.com/2012/09/12/measuring-prestige-analyzing-the-u-s-news-world-report-college-rankings/>
- <https://edu.srtfairs.com/beyond-university-ranking-what-makes-an-institution-prestigious/>